Augmented Reality

<u>Lab 6 - Report</u>

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The goal of this lab was to create an application utilising the Conda environment, OpenCV, and MediaPipe to detect faces present in input data from a webcam. Eventually the code needed to be altered to use public video footage as input data and then output the annotated video file.

Camera Playback

The face landmark detection application operated as intended. When using the playback from my laptop's camera, it was able to detect key points on my face and generate face landmarks. Changing facial expressions and tilting my head in different directions showed the face landmarks adapt accordingly. Obstructing the camera's view of my face in any way would remove the landmarks.

Video Playback

The input video is of a scene from a live-action TV show. It contains two characters conversing with each other, with the camera changing view to whoever is currently talking. While the face landmark detection application operated as intended for the first character, even going as far as to show their precise eye movements, it was observed that the application could not properly detect the key points of the second character’s face, which could be due to the character's glasses obstructing their face, the shading and distance in the shot, or the video quality.